7. A respirometer is a device used to measure the rate of respiration of a living organism by measuring its rate of exchange of oxygen and/or carbon dioxide. A simple respirometer used in a school laboratory is shown below. It can be used to estimate the volume of carbon dioxide in exhaled air.

Fact file

- Bromothymol blue indicator is green when neutral and blue when alkali.
- 5.6 cm^3 of carbon dioxide will neutralise the sodium hydroxide in the respirometer.

The following equation can be used to calculate the volume of carbon dioxide produced per minute.

Volume of carbon dioxide per minute = breathing rate × volume of carbon dioxide in one breath
(cm^3/minute) (breaths/minute) (cm^3/breath)

Examiner only

Use the information from the fact file opposite to answer the following questions.

(a) (i) Alun wanted to investigate the effect of exercise on the volume of carbon dioxide breathed out. While resting, he takes 5 breaths over a period of 25 seconds to change the colour of the Bromothymol indicator from blue to green. Use the equation given in the fact file to calculate the volume of carbon dioxide produced per minute. [3]

volume of carbon dioxide per minute = ... cm^3/minute

(ii) Suggest a possible source of error in the experimental method that could lead to inaccurate results. [1]

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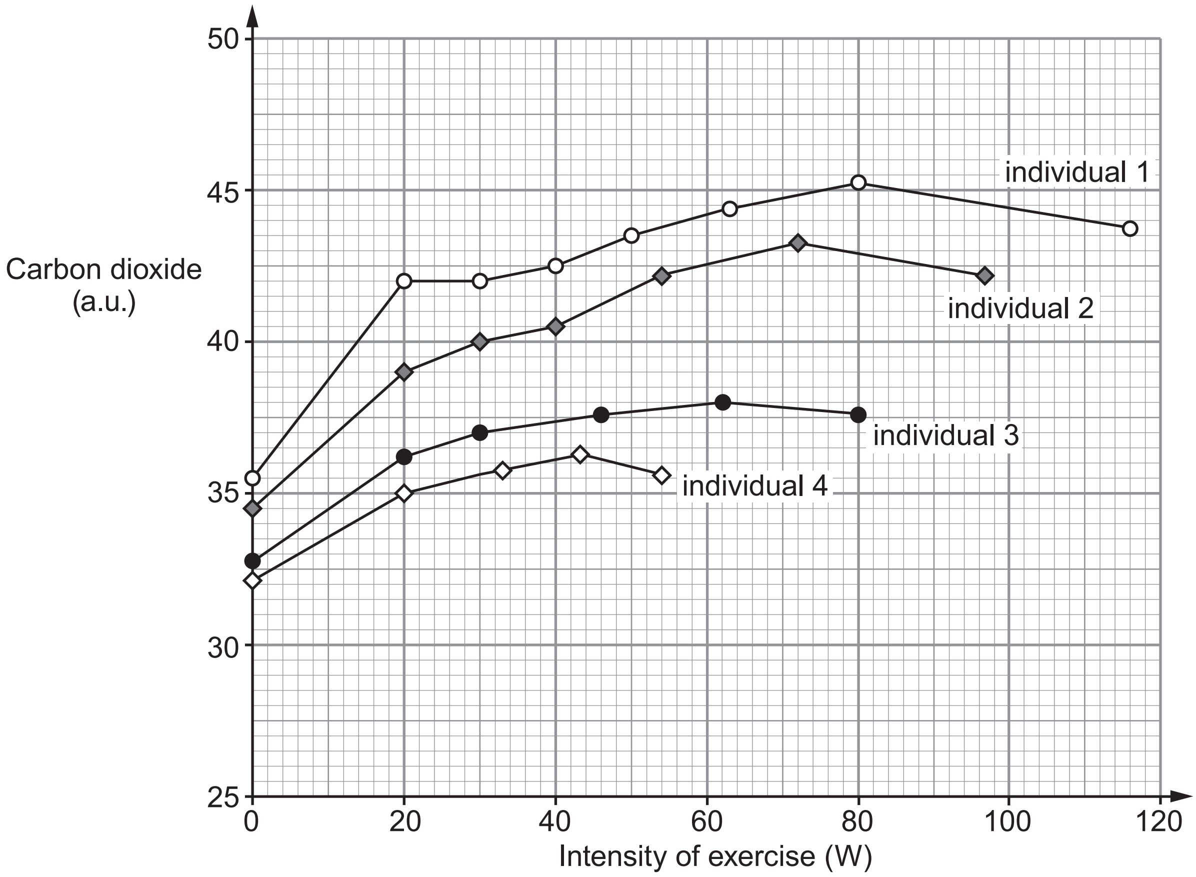
....

Examiner only

(b) A group of scientists monitored the proportion of carbon dioxide expired in 4 individuals. The individuals were tested on an exercise bike. They were allowed to rest for 4 minutes, and then exercised at an intensity of 20 watts for 4 minutes. After this time the intensity was increased 1 watt every 6 seconds. Individuals stopped exercising when they suffered cramp in their leg muscles. A gas analyser was used to get an accurate measurement of the expired carbon dioxide level per breath.

The results of the investigation are shown below.



(i) Explain the increase in carbon dioxide levels between an intensity of exercise 0 W and 20 W for all the individuals. [2]

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Examiner only

(ii) The decrease in carbon dioxide levels for all individuals at a higher intensity of exercise was linked to cramp in the muscles. Suggest an explanation for this. [2]

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(iii) Suggest which individual is the least fit. Give a reason for your answer. [1]

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....

(c) State why sports scientists working with athletes would measure oxygen consumption as well as carbon dioxide production. [1]

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